

第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

$$\frac{\Delta y}{\Delta x} = \frac{f(-1) - f(2)}{-1 - 2} = \frac{1 - 2}{-3} = \frac{1}{3}$$

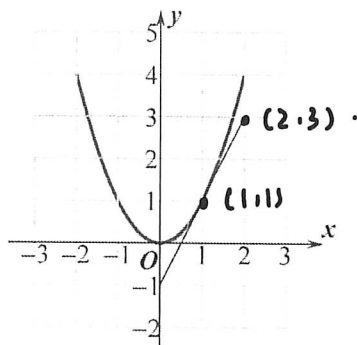
1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

(A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3

2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$? $f'(0) = \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x) - f(0)}{\Delta x}$

(A) 4 (B) -4 (C) -3 (D) 4

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) =$ (D) $= 3$



(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

(A) $\frac{5}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$

5. **Calculator** Let $y = \tan e^x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5}$

(A) -12.056 (B) 0 (C) 271.066 (D) 273.066

6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.354 (B) -0.001 (C) -2.666 (D) 0

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

- (A) At time $t = 10$, the population of insects is 230.
 (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t = 10$, the population of insects is ~~decreasing~~ at a rate of 230 insects per day.
 (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 ✓ (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$



Free Response Questions

9. Let $f(x) = x^2 - 2x$.

(a) Find the average rate of change of f over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x = 2$.

$$(a) \frac{\Delta y}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{-4}{-2} = 2$$

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$(b) \frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 2(2+h) - 0}{h} = h + 2$$

$$= 2$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\begin{aligned} f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cancel{x^3} + 3x^2\Delta x + 3x(\Delta x)^2 + \cancel{\Delta x^3} - x^3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{3x^2\Delta x + 3x(\Delta x)^2}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} 3x^2 + 3x\Delta x \\ &= 3x^2 \\ f'(2) &= 3 \times 4 = 12 \end{aligned}$$